

ANINDYA MONDAL

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EDUCATION

University of Surrey

Guildford, UK

PhD Candidate in Artificial Intelligence

Oct 2022 – Present

- Institute: Surrey Institute for People-Centred AI, CVSSP.
- Research focus: Integrating auxiliary information for natural world understanding, with emphasis on multimodal learning, vision-language integration, and generative AI.

Jadavpur University

Kolkata, India

B.E. (Hons.) in Electronics & Telecommunication Engineering

Aug 2018 – Jun 2022

- GPA: 8.79/10

RESEARCH FOCUS

- 2023–2025: Vision–language integration, multimodal learning, action recognition, text-to-image synthesis, object counting, 3D/4D generation.
- 2020–2022: Graph neural networks, time-series analysis, graph signal processing, neuromorphic vision, subspace learning.

RESEARCH EXPERIENCE

University of Surrey

Guildford, UK

Doctoral Researcher

Oct 2022 – Present

- Developed **CountLoop**, an LLM agent-guided, training-free diffusion framework for high-instance text-to-image generation.
- Created **OmniCount**, a class-agnostic object counting model using semantic and geometric priors.
- Proposed **MSQNet**, a transformer-based multimodal action-recognition model (50% precision gain).
- Led benchmark creation for animal action recognition, detection, and segmentation.

Indian Institute of Science

Bengaluru, India

Research Intern

May 2022 – Aug 2022

- Devised a source-free domain-adaptation framework for image classification, enhancing robustness.

Jadavpur University

Kolkata, India

Undergraduate Research Assistant

Oct 2020 – May 2022

- Implemented a Sobolev-norm minimisation algorithm for time-varying graph-signal reconstruction.
- Developed a graph-based semi-supervised learning framework for semantic segmentation.
- Improved moving-object detection from event data using graph spectral clustering.

PUBLICATIONS

CountLoop: Iterative Agent Guided High Instance Image Generation.

A. Mondal, A. Banerjee, S. Nag, J. Lladós, X. Zhu, A. Dutta. Preprint. [Project]

OmniCount: Multi-label Object Counting with Semantic–Geometric Priors.

A. Mondal, S. Nag, X. Zhu, A. Dutta. AAAI 2025. DOI: 10.1609/aaai.v39i18.34151

Actor-Agnostic Multi-label Action Recognition with Multi-modal Query.

A. Mondal, S. Nag, J.M. Prada, X. Zhu, A. Dutta. ICCVW 2023. DOI: 10.1109/ICCVW60793.2023.00086

Time-Varying Signals Recovery via Graph Neural Networks.

J.A.C. Correa, J.H. Giraldo, A. Mondal, *et al.* ICASSP 2023. DOI: 10.1109/ICASSP49357.2023.10096168

Recovery of Missing Sensor Data by Reconstructing Time-Varying Graph Signals.

A. Mondal, *et al.* EUSIPCO 2022. DOI: 10.23919/EUSIPCO55093.2022.9909940

Moving Object Detection for Event-based Vision using Graph Spectral Clustering.

A. Mondal, R. Shashant, *et al.* ICCVW 2021. DOI: 10.1109/ICCVW54120.2021.00103

TECHNICAL SKILLS

- **Languages:** Python (expert), MATLAB (expert), C (advanced).
- **Libraries/Frameworks:** PyTorch (expert), scikit-learn (advanced), NumPy, SciPy, Pandas.
- **Tools:** Git, Docker, Jupyter Notebook, L^AT_EX.

AWARDS AND HONOURS

AAAI 2025 Conference & Travel Grant, Philadelphia, USA.

ICCV 2023 Conference Grant, Paris, France.

Postgraduate Studentship, University of Surrey (2022).

Uplink Research Internship Award, ACM SIGKDD India Chapter (2022).

PROFESSIONAL EXPERIENCE

University of Surrey

Guildford, UK

Teaching Assistant

2023 – 2025

- Applied Machine Learning (EEEM068).
- Advanced Topics in Computer Vision & Deep Learning (EEEM071).

Reviewer

Conferences & Journals

2022 – 2025

- Reviewer for CVPR, ICCV, ECCV, NeurIPS, ICASSP, BMVC, IEEE TSP, TSIPN, ISBI.